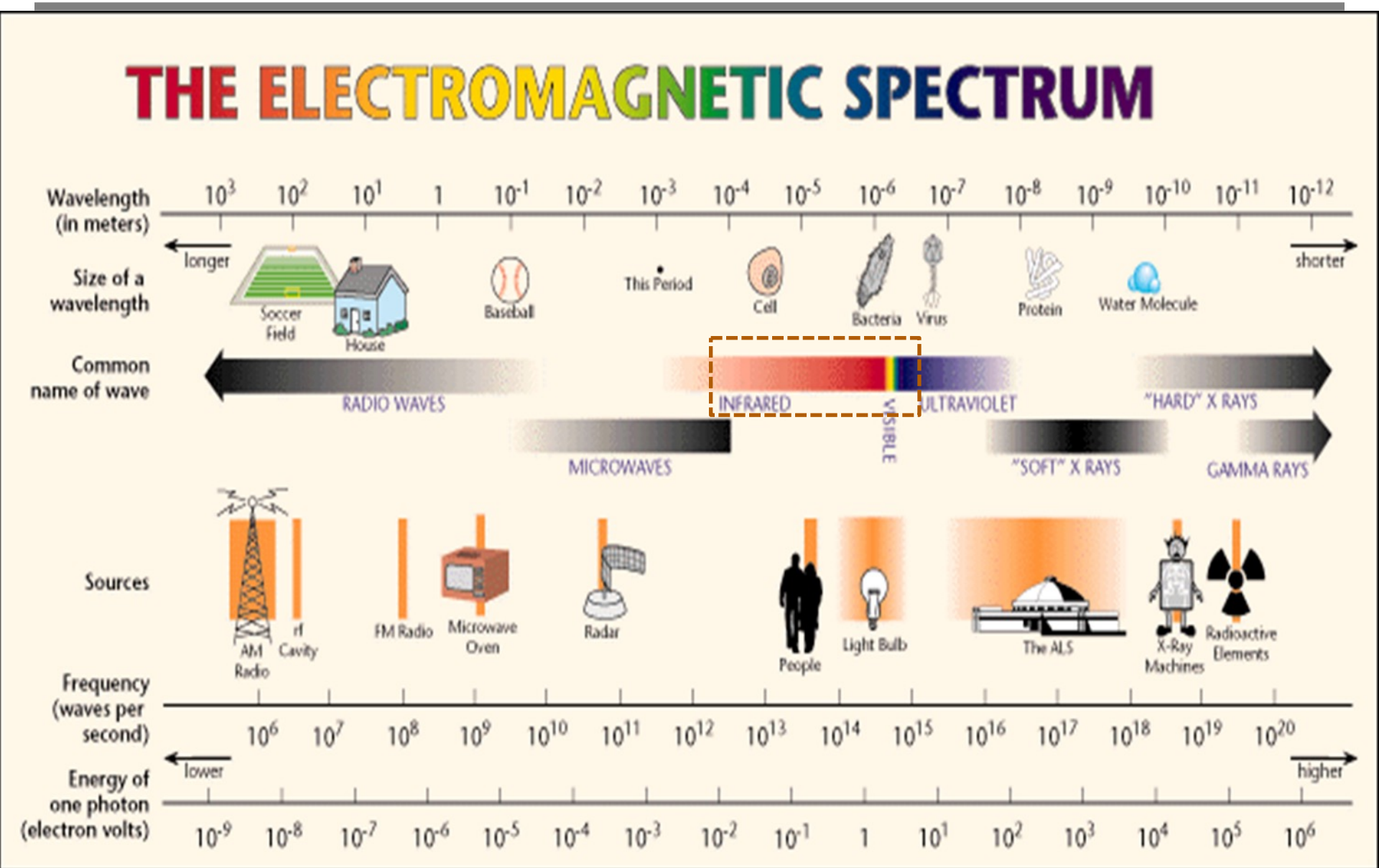
Light as an Electromagnetic Wave

The following subjects will be covered

- More Examples on Light Wave Propagation in Open Space.

Optical Spectrum



Example

The visible light complex refractive index curve of GaAs is shown in the figure. For red light (680 nm), find (a) the value of the complex refractive index, (b) the speed of light and (c) the attenuation coefficient (d) How far would a red light beam travel inside a GaAs material before it reduces to 1% of its original power density value?

$$n = 3.1826 - j0.18$$

Speed of light

$$v = \frac{c}{n'} = 0.943\times10^{8} \quad (\mathrm{m/s})$$

The attenuation coefficient is

$$\alpha = \frac{\omega}{c} n'' = 1.5\times10^{6} \quad (1/\mathrm{m})$$

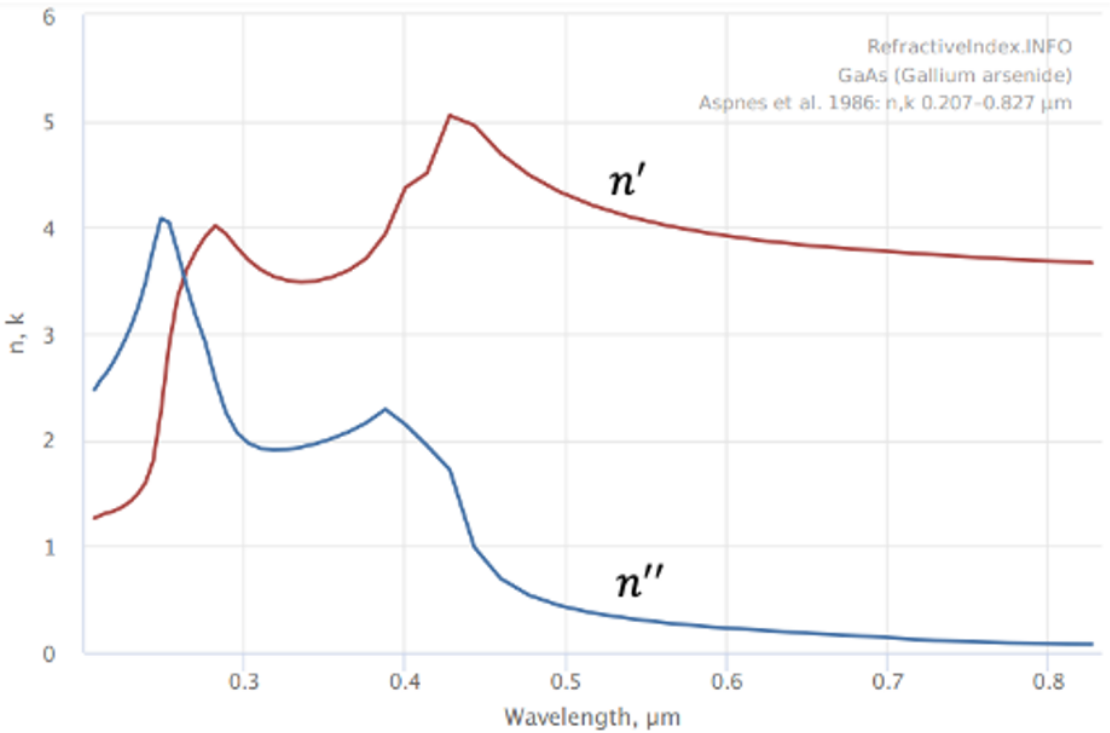
The absorption coefficient is

$$\alpha = 2\frac{\omega}{c} n'' = 3\times10^{6} \quad (1/\mathrm{m})$$

Find the distance using

$$0.01 = e^{-3\times10^{6} z}$$

$$z = 1.54\ \mu m$$



https://refractiveindex.info/

Examples

1) Discuss the wave direction. $\vec{E} = E_o \vec{a}_y \, e^{j\omega t} \, e^{-j(x+2z)/\sqrt{2}}$ (V/m)

Transverse Electric (TE)

$k_x = 1/\sqrt{2}$ $k_z = 2/\sqrt{2}$ $\vec{k} = \frac{1}{\sqrt{2}} \vec{a}_x + \frac{2}{\sqrt{2}} \vec{a}_z$ (rad/m)

2) The complex electric field intensity of a time-harmonic wave travelling in a source-free free space is given by (distance in meters)

$$\vec{E} = (80 \, \vec{a}_x - 60 \, \vec{a}_z) \, e^{-j(6x+8z)} \text{ (V/m)}$$

What is the wavelength in the wave propagation direction?

$k_x = 6$ $k_z = 8$ $k = \sqrt{6^2 + 8^2} = 10$ (rad/m) $\lambda = \frac{2\pi}{k} = 0.628$ (m)

What is the angle the wave propagation direction makes with z?

$\tan\theta = 6/8$ $\theta = 36.9°$

Transverse Magnetic (TM)

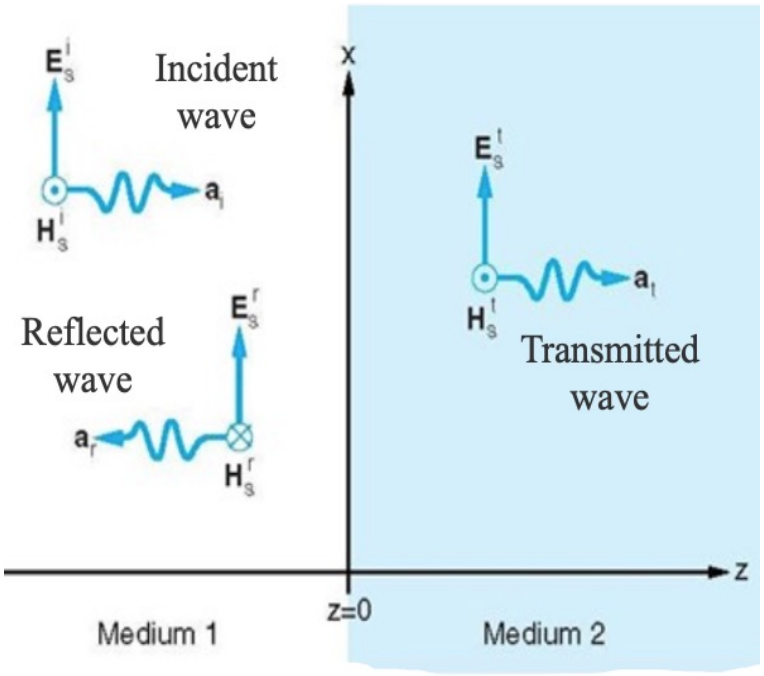
Light as an Electromagnetic Wave

Part II: Light Behavior and Properties

The following subjects will be covered

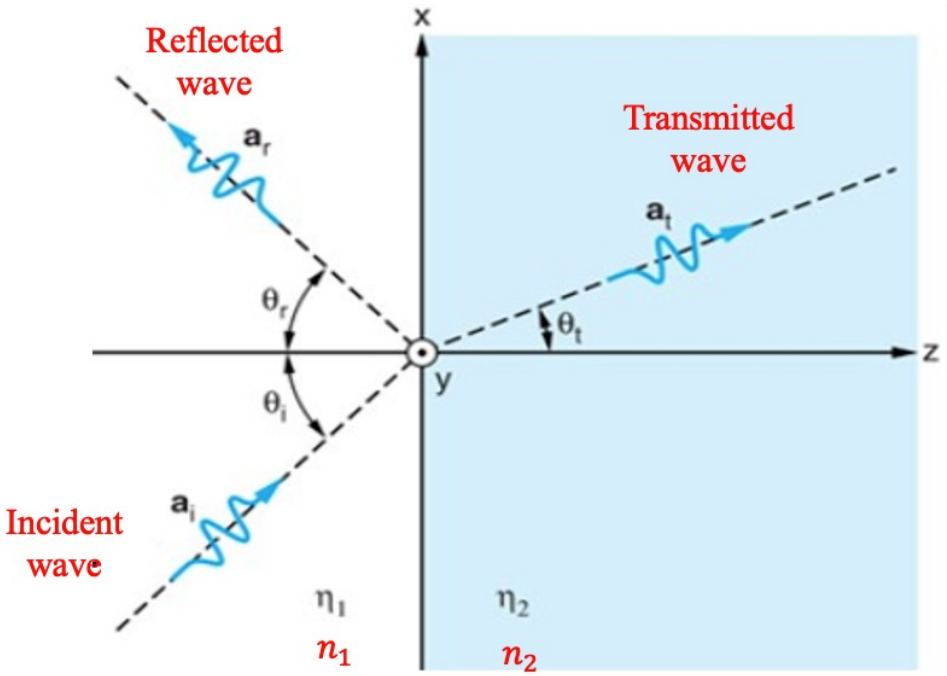
- Reflection, Refraction and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

Light Reflection and Transmission



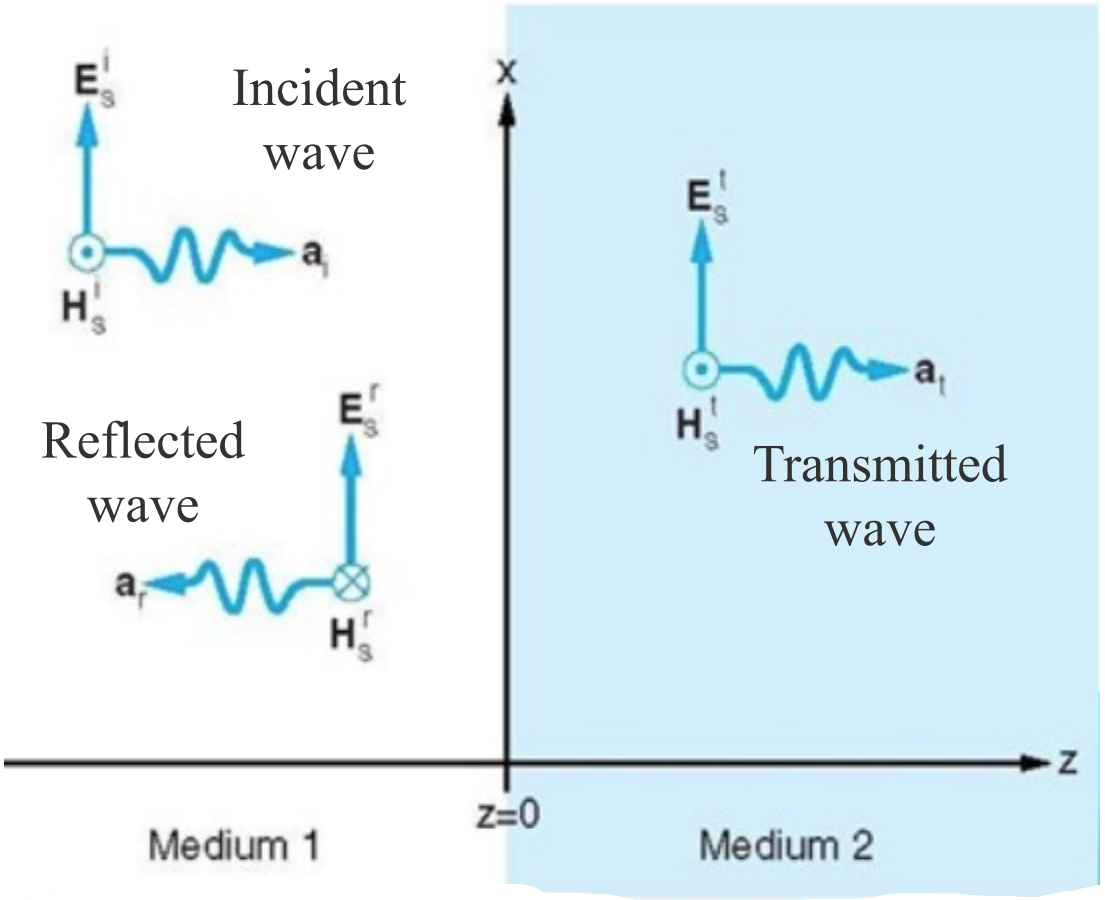
n_1, k_1, α_1 n_2, k_2, α_2

Normal Incidence



Oblique Incidence

Single Interface: Normal Incidence



n_1, k_1, α_1 n_2, k_2, α_2

Single Interface: Normal Incidence

Incident Wave

$$\vec{E}_i(z) = E_o\, e^{-jk_1 z}\, \vec{a}_x$$

$$\vec{H}_i(z) = \frac{E_o}{\eta_1} e^{-jk_1 z}\, \vec{a}_y$$

Reflected Wave

$$\vec{E}_r(z) = \Gamma\, E_o\, e^{+jk_1 z}\, \vec{a}_x$$

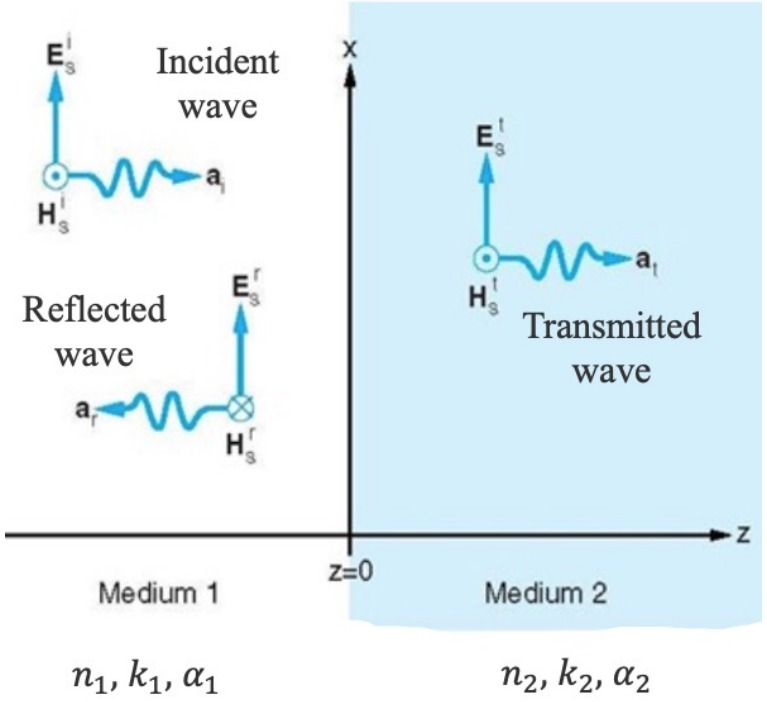
$$\vec{H}_r(z) = \frac{\Gamma E_o}{\eta_1}\, e^{+jk_1 z}\ \ (-\vec{a}_y)$$

Transmitted Wave

$$\vec{E}_t(z) = \tau E_o\, e^{-jk_2 z}\ \ \vec{a}_x$$

$$\vec{H}_t(z) = \frac{\tau\, E_o}{\eta_2}\, e^{-jk_2 z}\ \ \overrightarrow{a_y}$$

$$\eta_1 = \frac{\eta_o}{n_1} \qquad \eta_2 = \frac{\eta_o}{n_2} \qquad \eta_o = \sqrt{\mu_o/\varepsilon_o}$$



n_1, k_1, α_1 $\qquad$ n_2, k_2, α_2

Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

Single Interface: Normal Incidence

Apply boundary conditions

$$E_{1_tan} = E_o(1+\Gamma)$$

$$E_{2_tan} = E_o\tau$$

$$H_{1_tan} = \frac{E_o}{\eta_1}(1-\Gamma)$$

$$H_{2_tan} = \frac{\tau E_o}{\eta_2}$$

Field Coefficients

Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\Gamma = \tau - 1$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

Power Coefficients

Reflectance

$$R = |\Gamma|^2$$

$$R + T = 1$$

Transmittance

$$T = \frac{n_2}{n_1}|\tau|^2$$

Power Conservation

1) Γ and τ are in general complex.

2) $0 \le |\Gamma| \le 1$

$$\vec{S}^i_{ave} = \frac{n_1|E_o|^2}{2\eta_o}\,\vec{a}_k$$

$$\vec{S}^t_{ave} = n_1 T\frac{|E_o|^2}{2\eta_o}\,\vec{a}_k$$

$$\vec{S}^r_{ave} = n_1 R\frac{|E_o|^2}{2\eta_o}\,(-\vec{a}_k)$$

Single Interface: Normal Incidence

Reflectance (Power)

$$R = |\Gamma|^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$$

Transmittance (Power)

$$T = 1 - R = \frac{4\, n_1 n_2}{(n_1 + n_2)^2}$$

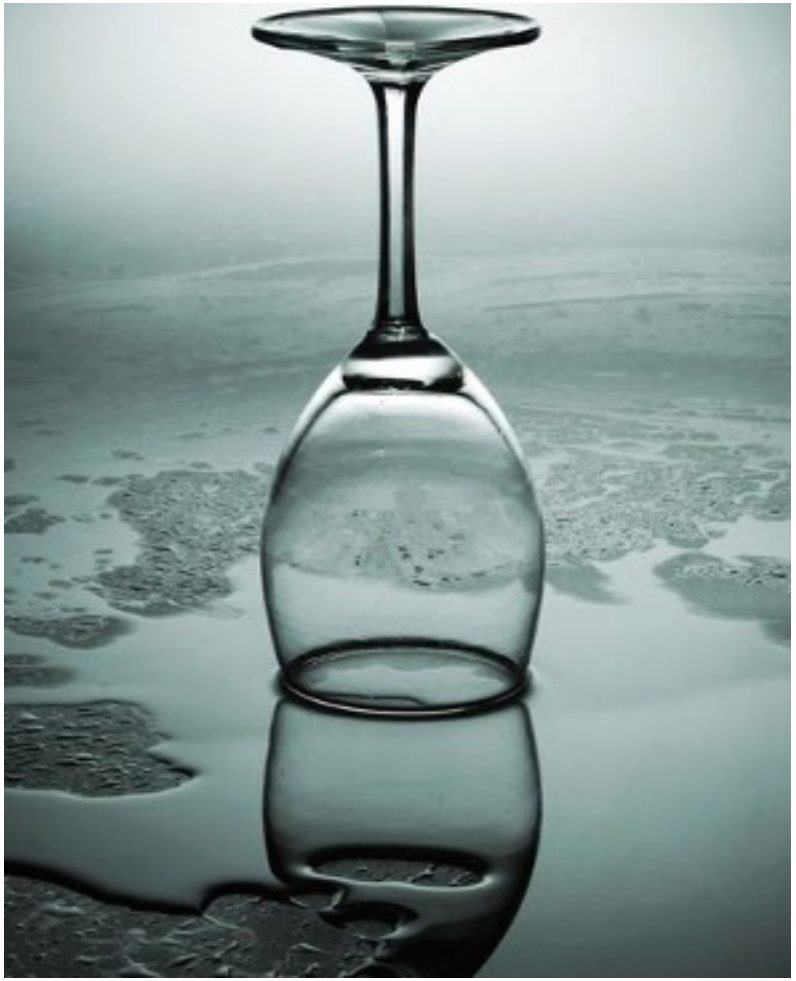
Example: Glass Reflection

Glass refractive index, $n \approx 1.52$

$$\Gamma = \frac{1.0-1.52}{1.0+1.52} = -0.2065$$

$$R = (-0.2065)^2 = 0.0426$$

$$T = 1 - 0.0426 = 0.957$$



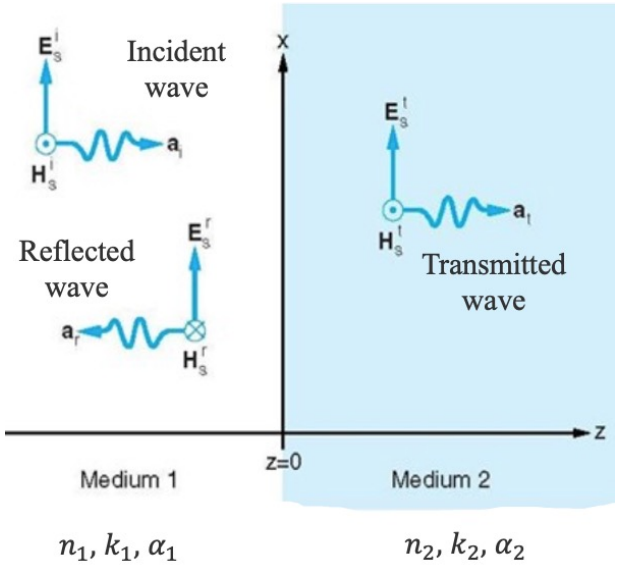
96% of light power goes through!

Remarks

1) Medium 1 perfect dielectric (n_1=real) and medium 2 perfect conductor ($n_2 \to \infty$)

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = -1$$ Total reflection

$$\tau = 0$$ Zero transmission



2) Medium 1 is a denser dielectric than medium 2

$$n_1 > n_2 \qquad \Gamma > 0$$

3) Medium 1 is a lighter dielectric than medium 2

$$n_1 < n_2 \qquad \Gamma < 0$$

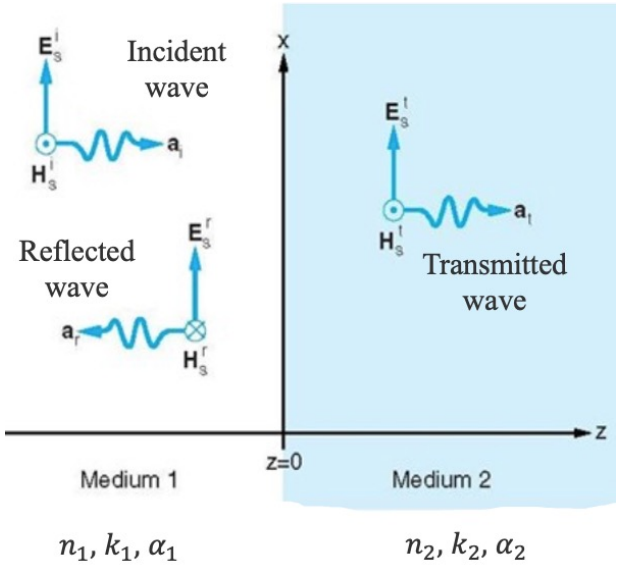
Lossy Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

(n_1 and n_2 complex quantities)

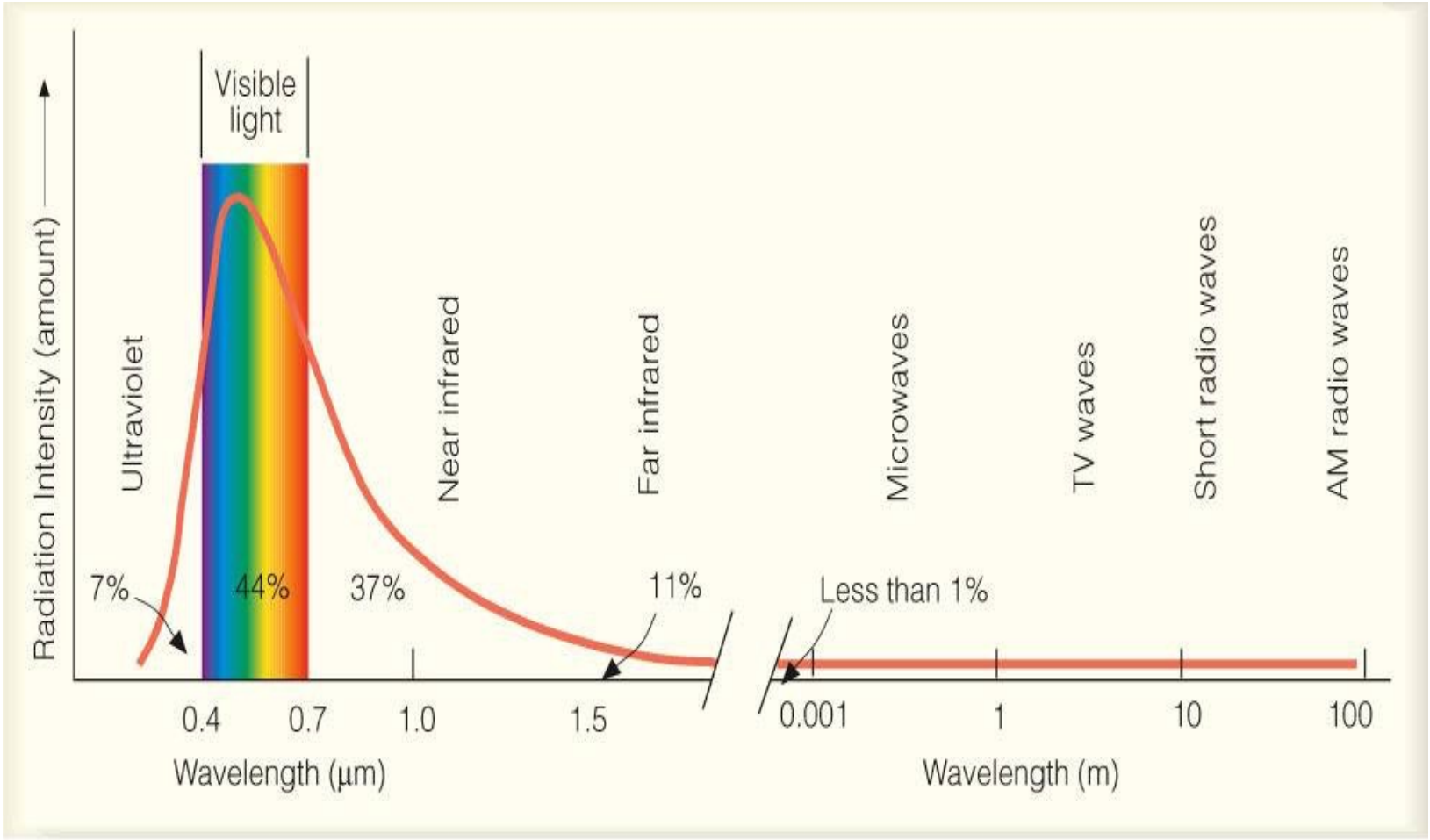
$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, \vec{a}_z$$

$$\vec{S}^r_{ave} = |\Gamma|^2\, \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2\, \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right)\, \vec{a}_z$$

Example: Solar Cell Design Considerations

It is required to select between Si and Ge for the design of a solar cell. The selection is based on the amount of reflection at normal incidence.



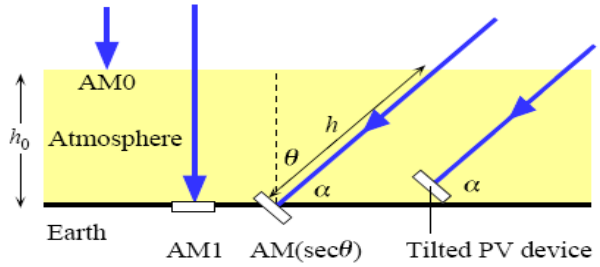
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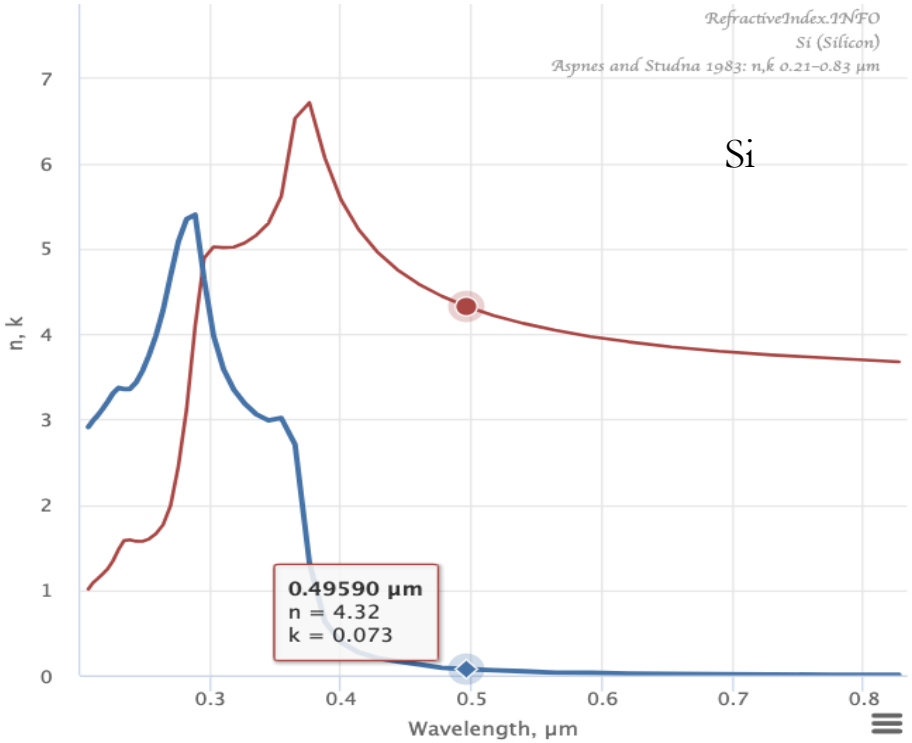
Example: Solar Cell Design Considerations

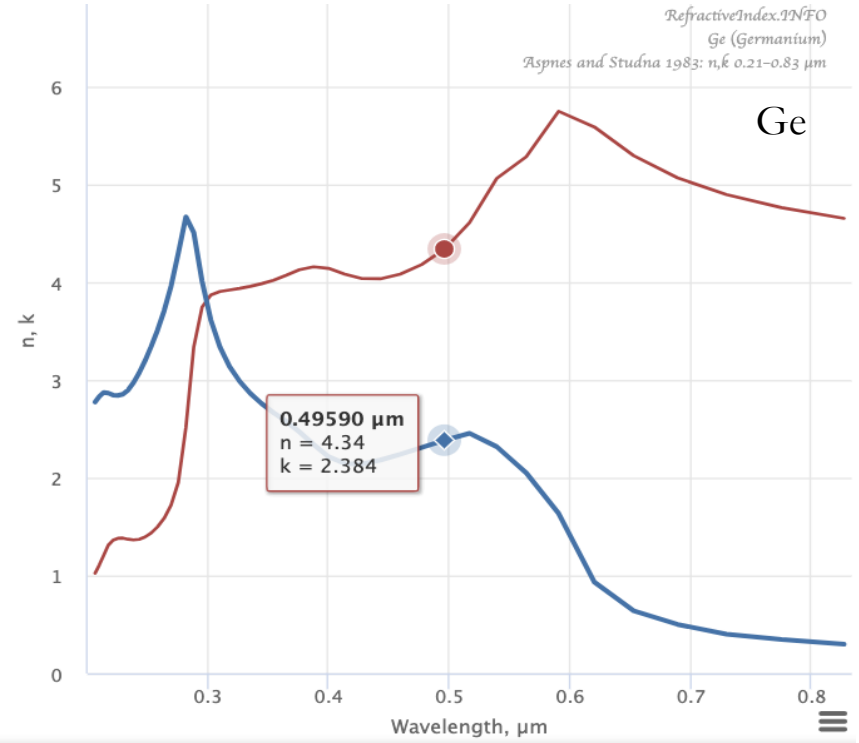
Refractive index curves for Si and Ge. Note, in the figures n and k are n' and n''.

Si (0.5 μm): $n = 4.32 - j0.073$.

Ge (0.5 μm): $n = 4.34 - j2.384$.





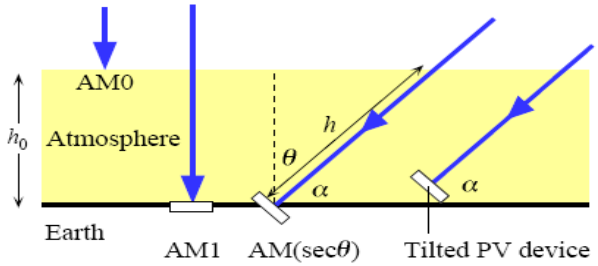


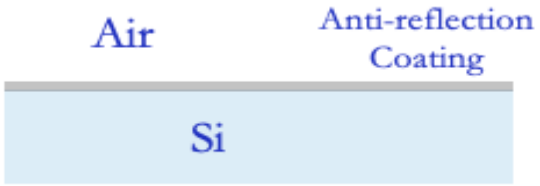
Example: Solar Cell Design Considerations

Refractive index curves for Si and Ge. Note, in the figures n and k are n' and n''.

Si (0.5 m): $n = 4.32 - j0.073$.

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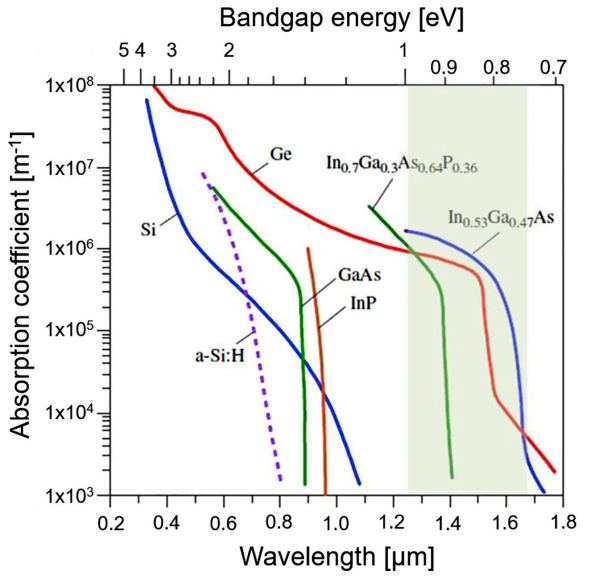


$$\Gamma_{Si} = \frac{1.0 - (4.32 - j0.073)}{1.0 + (4.32 - j0.073)} = 0.6184 \angle 170.9°$$

$$R_{Si} = 0.382$$

$$\Gamma_{Ge} = \frac{1.0 - (4.34 - j2.384)}{1.0 + (4.34 - j2.384)} = 0.7017 \angle 120.4°$$

$$R_{Ge} = 0.492$$



Q & A

Send me an email if you have any questions.